NMC 2013 PRIMARY LEVEL PAPER 2 QUESTIONS

SECTION A : 5 marks each

Qn. 1. Given that $x = \frac{1}{4}$. Which one of $x, x^2, \frac{x}{2}$ and $\sqrt{x}$ has the largest value?

Qn. 2. If the mean, median and mode of $7,\ 7,\ 5,\ 7,\ x$ are the same, what is the value of x?

Qn. 3. What is 50% of 2013 plus 2013% of 50?

Qn. 4. A father said to his son "I was as old as you now at the time of your birth." If the father's age is 38 years now, find the son's age five years back.

Qn. 5. If one-third of one-fourth of a number is 15. what is then three-tenth of the number?

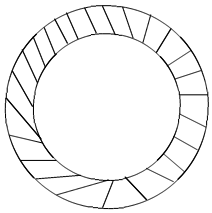
Qn. 6. An aeroplane covers a certain distance at a speed of 240kmh^{-1} in 5hours. To cover the same distance in $1\frac{2}{3}$hours. At what speed must the aeroplane fly?

Qn. 7. Given that a and b are odd numbers, find the odd man out: $ab,\ a+b+1,\ 2a+3b+7,\ 2a+b$.

Qn. 8. Find the sum of five consecutive integers if the middle one is n?

Qn. 9. If the length of the side of a square is doubled, what is the ratio of the areas of the original square to the area of the new square?

Qn. 10. A circular garden with a diameter of 20metres is surrounded by a walkway (shaded part) of width 1metre. Find the area of the walkway. (Take $\pi = \frac{22}{7}$)



SECTION B : 10 **marks each**

Qn. 11. Find x if $\frac{x}{\sqrt{128}} = \frac{\sqrt{162}}{x}$

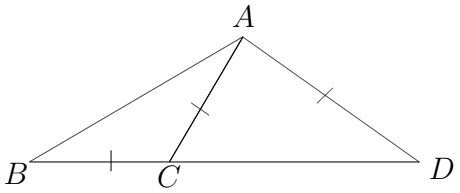
Qn. 12. A train can travel 50% faster than a car. Both start from point A at the same time and reach point B, 75km away from A at the same time. On the way, however, the train lost about 25 minutes while stopping at the stations. Find the speed of the car.

Qn. 13. The ratio of the perimeter to the width of a rectangle is 5 : 1. If the area of the rectangle is 24cm^2,

(a) What is the length of the rectangle?

(b) Calculate the perimeter of the rectangle.

Qn. 14. In triangle ABD, C is a point on BD such that $BC = CA = AD$ and angle $BAC = 20^o$. Find angle BAD.



Qn. 15. Nine numbers are written in ascending order. The middle number is also the average of the nine numbers. The average of the five largest numbers is 68 and the average of the five smallest numbers is 44. What is the value of the middle number?

NMC 2013 PRIMARY LEVEL PAPER 2 SOLUTIONS

SECTION A : 5 marks each

Qn. 1. $x = \frac{1}{4}$, $x^2 = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$, $\frac{x}{2} = \frac{1}{4} \div 2 = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$, $\sqrt{x} = \sqrt{\frac{1}{4}} = \frac{1}{2}$.

Hence, the largest value is $\sqrt{x} = \frac{1}{2}$

Qn. 2. Since 7 appears three times out of five numbers, then mode= 7 regardless of the value of x.
Since mean= mode= median=7, $\frac{7+7+5+7+x}{5} = 7 \Longrightarrow 26 + x = 35$ so that $x = 9$.

Qn. 3. $\frac{50}{100} \times 2013 + \frac{2013}{100} \times 50 = \frac{1}{2} \times 2013 + \frac{1}{2} \times 2013 = \frac{2013}{2} + \frac{2013}{2} = 2013$.

Qn. 4. Let the son's present age be x years. Then, $38 - x = x \Longrightarrow x = 19$.
The son's age five years back $= 19 - 5 = 14$ years.

Qn. 5. Let the number be x. Then, $\left(\frac{1}{3}\right)\left(\frac{1}{4}\right) x = 15 \Longrightarrow \frac{1}{12}x = 15$ so that $x = 12 \times 15 = 180$. Therefore, $\frac{3}{10} \times x = \frac{3}{10} \times 180 = \frac{540}{10} = 54$.

Qn. 6. Distance = speed $\times$ time $= 240 \times 5 = 1200km$.

$$\text{Speed} = \frac{\text{Distance}}{\text{time}} = \frac{1200}{1\frac{2}{3}} = 1200 \times \frac{3}{5} = 720kmh^{-1}.$$

Qn. 7. ab is an odd number being a product of odd numbers.
$a + b + 1$ is an odd number since when 1 added to an even number gives an odd number.
$2a + 3b + 7$ is an even number since it is a sum of an even number and two odd numbers.
$2a + b$ is a sum of an even number and an odd number which is an odd number.
Therefore an even number is the odd man out and this is $2a + 3b + 7$.

Qn. 8. The five consecutive integers whose middle one is n are $n - 2, n - 1, n, n + 1$ and $n + 2$.
The sum of these integers is $n - 2 + n - 1 + n + n + 1 + n + 2 = 5n$.

Qn. 9. Area of the original square $= x \times x = x^2$.
Area of the new square $= 2x \times 2x = 4x^2$.
Ratio of original area to new area $= x^2 : 4x^2 = 1 : 4$.

Qn. 10. Area of the garden $= \pi r^2 = \pi(10)^2$.
Area of the garden + walkway $= \pi R^2 = \pi(11)^2$
Area of the walkway= $\pi R^2 - \pi r^2 = \pi(11)^2 - \pi(10)^2 = 121\pi - 100\pi = 21\pi = 21 \times \frac{22}{7} = 66m^2$.

SECTION B : 10 marks each

Qn. 11. $\frac{x}{\sqrt{128}} = \frac{\sqrt{162}}{x}$
By cross multiplication we get

$x \times x = \sqrt{128} \times \sqrt{162} = \sqrt{128 \times 162} = \sqrt{64 \times 2 \times 18 \times 9} = \sqrt{64 \times 36 \times 9} = \sqrt{8^2 \times 6^2 \times 3^2} = 8 \times 6 \times 3 = 1$

Taking square root both sides gives $x = \sqrt{144} = 12$.

Qn. 12. Let the speed of the car be x. This implies that the speed of the train $= 1\frac{1}{2}x = \frac{3}{2}xkmh^{-1}$.
Time (t) taken by the car to cover $75km = \frac{D}{S} = \frac{75}{x}$ hrs.
The train covers the journey in $thrs - 25min = (t - \frac{25}{60})hrs$.
Therefore, $t - \frac{25}{60} = \frac{D}{s} = \frac{75}{\frac{3}{2}x}hrs$ so that $t - \frac{25}{60} = \frac{75 \times 2}{3x} = \frac{150}{3x} = \frac{50}{x}$.

Since $t = \frac{75}{x}$ (car time), this implies that $\frac{75}{x} - \frac{25}{x} = \frac{50}{x}$.
multiply every where by x, to get $75 - \frac{25}{60}x = 50$ and solving this equation gives $x = 60$.
Therefore the speed of the car is $60kmh^{-1}$.

Qn. 13. (a) $\frac{p}{w} = \frac{2(l+w)}{w} = \frac{5}{1}$
$2l + 2w = 5w$ which implies that $2l = 5w - 2w = 3w \Longrightarrow w = \frac{2}{3}l$
But since $A = l \times w = 24$, we get $l \times \frac{2}{3}l = 24$
$2l^2 = 72 \Longrightarrow l^2 = 36$ and taking square root, we get $l = 6$.
The length of a rectangle is $6cm$.

(b) $w = \frac{2l}{w} = \frac{2 \times 6}{3} = 4$.
Perimeter $= 2(l + w) = 2(6 + 4) = 20cm$.

Qn. 14. In the triangle, since $\bar{BC} = \bar{CA}$, then $< ABC =< BAC = 20^o$
$< ABC + < BAC + < ACB = 180^o \Longrightarrow 20^o + 20^o + < ACB = 180^o$ so that $< ACB = 180^o - 40^o = 140^o$.
$< ACB + < ACD = 180^o \Longrightarrow 140^o + < ACD = 180^o \Longrightarrow < ACD = 40^o$. Since $\bar{AC} = \bar{AD}$, this implies $< ACD =< ADC = 40^o$.
$< ACD + < ADC + CAD = 180^o$
$40^o + 40^o + < CAD = 180^o$
$80^o + < CAD = 180^o$. Hence, $< CAD = 100^o$.
Therefore, $< BAD + < BAC + < CAD = 20^o + 100^o = 120^o$.

Qn. 15. Let the sum of all the numbers be S, so the middle number is $\frac{1}{9}S$. The sum of the five largest numbers is $5 \times 68 = 340$, and the sum of the five smallest numbers is $5 \times 44 = 220$. If we add these sums together, then we have counted the middle number twice, so $S + \frac{1}{9}S = 340 + 220$, that is $\frac{10}{9}S = 560$. Thus $S = \frac{560 \times 9}{10} = 504$. Therefore the middle number is $\frac{1}{9}S = \frac{1}{9} \times 504 = 56$.